MIMIC-IV stage: pre-specified analysis plan M1 v1.2

Laboratory draws before ward-to-ICU escalation, rebuilt on specimen_id with matched non-escalated controls. Written and tested on the open 100-patient MIMIC-IV demo before any run on the full data. To be frozen, then run once on the author's computer.

Why this analysis

The paper's real-data premise is that clinicians observe a patient more often as the patient deteriorates. The earlier reanalysis showed a rise in lab activity before ICU transfer that survived restriction to ward origins (2.15 to 3.78 tests/hour, n = 15,042), but three gaps remained: draws were approximated by a 60-second timestamp rule, there was no comparison group and no interval estimate, and time-of-day effects were not addressed. M1 closes all three.

Design

Element	Definition
Cases	First ICU entry in an admission whose preceding unit is a ward, with the 48 h before ICU entry spent entirely on ward units (no ED, procedural area, ICU or unknown unit). One admission per patient (earliest).
Controls	Risk-set matching on clock hour with an elapsed-time caliper. For each case (seeded random order), candidate ward admissions with no ICU entry are examined in seeded random order. The pseudo-event is the time with the case's exact clock hour and minute nearest to (candidate admission + the case's hours since admission), so the elapsed-time difference is at most ±12 h by construction. It is accepted only if the 48 h before it lie within one continuous ward run. Up to 2 controls per case, without replacement (each control patient used once; patients who are ever cases are excluded). Selection uses transfer records only, never laboratory data.
Unmatched cases	Counted and reported, never silently dropped. The case–control contrast uses matched sets (a case with its controls); cases without an eligible control are reported separately.
Windows	Baseline 48–24 h before the event; pre-event 24–0 h. Each window covers all 24 clock hours.
Time of day	Built in twice: both windows span the full day (morning-rounds draws fall equally in each), and controls are matched on the event's clock hour. The 12 h sensitivity analysis relies on the clock-hour matching.
Primary outcome	Draws per hour = distinct specimen_id.
Secondary	Distinct collection times per hour (one venipuncture can yield several specimen_ids); tests per hour; tests per draw; the old 60-second rule and its agreement with specimen_id.

Element	Definition
Primary estimate	Equal-set-weighted difference-in-differences of draws/hour. For each matched set: (case pre − case baseline) − mean over that set's controls of (control pre − control baseline). The estimate is the average of these set contrasts, each set weighted equally whether it has 1 or 2 controls. Control means and ratios are likewise averages of set-level control means. 95% interval: bootstrap of matched sets (2,000 resamples, seed 20261101), recomputing the same estimand in every resample. One admission per patient throughout.
Matching diagnostics	Reported for every run: elapsed-time difference for all matched pairs (mean, median, range, maximum absolute, quantiles), share with exact clock-hour-and-minute match, share of cases and controls with 48 h continuous ward occupancy verified independently, one admission per patient, no patient both case and control, and cases with 2, 1 or 0 controls.
Also reported	Case ratio pre/baseline, control ratio, ratio of ratios (all with intervals); share of patients with an increase; 48-hour profile of draws before the event; draws by clock hour; ED-origin escalations descriptively for contrast.
Sensitivity	12 h windows (12–24 h vs 0–12 h before the event).

Changes from v1.0

v1.0 placed the control pseudo-event at the case's elapsed time and then shifted it by up to 23 h to match the clock hour, which broke the elapsed-time match. Exact matching on both is not possible in general (two admissions that start at different times of day cannot reach the same clock hour at the same elapsed time), so v1.1 uses risk-set matching on exact clock hour with a ±12 h elapsed-time caliper, the smallest caliper that always permits an exact clock match, and resamples matched sets in the bootstrap. The outcome, windows, seed, bootstrap size and interpretation rule are unchanged. The change was made to meet the written specification, not in response to any outcome; the demo outcome is a code check and not evidence. Manuscript wording: “risk-set matching on clock hour with elapsed-time caliper matching (±12 hours)”.

Change from v1.1

v1.1 pooled all controls when computing control means, so a matched set with two controls carried twice the weight of a set with one. v1.2 makes every estimate equal-set-weighted, as defined above, and the bootstrap recomputes that same estimand. A built-in self-test (python mimic_specimen_analysis.py --selftest) checks the calculation on a hand-worked example with one 2-control set and one 1-control set (correct answer 2.75; pooled answer 2.0) and confirms the bootstrap recomputation. All other definitions are unchanged.

Interpretation rule (fixed now)

- **Supported:** the difference-in-differences interval for draws/hour lies above zero. The paper may then say that sampling density rises before clinician-initiated escalation, beyond what is seen in comparable ward patients at the same time of day and point in the stay.
- **Not supported:** the interval includes or lies below zero. The paper then reports that the rise is not distinguishable from background patterns and narrows its real-data claim accordingly.
- Whatever the result, it is reported as found. This analysis shows only that sampling density rises before escalation; it does not by itself show that this degrades early-warning signals (the simulations address that).

Privacy and data handling

- The script runs only on the author's computer and writes one file of aggregates: no patient rows, identifiers or dates.
- Any patient count below 10 is suppressed in the output.
- Only that summary file leaves the author's computer. Full-MIMIC data files are not uploaded to any online service, consistent with the PhysioNet credentialed data use agreement and its guidance on online services.

Demo test (open data, code check only)

On the 100-patient demo (v1.2): 11 ward-origin escalations; 10 matched with 2 controls each (20 controls), 1 with no eligible control (reported). Clock hour and minute matched exactly in 100% of pairs; elapsed-time difference median 1.4 h, range −11.8 to +11.9 h (all within the caliper); 48 h continuous ward occupancy verified independently for 100% of cases and controls; one admission per patient; no patient is both case and control. In the demo every matched set has 2 controls, so equal-set and pooled weighting coincide; the self-test covers the unequal case. Demo outcome numbers are far too small to mean anything and are not results. The 60-second rule finds about half as many draws as specimen_id, confirming that the definition matters.

ICU vitals

Not part of M1. The earlier vitals check found heart-rate cadence mostly clock-driven (92% of timestamps on the hour) and near-chance discrimination for intubation (AUC 0.51). It is reported as a documented feasibility result, not as evidence for the mechanism.